



AJAY PRASATH P 2024-CSE ▾

A2**Started on** Wednesday, 29 October 2025, 11:20 AM**State** Finished**Completed on** Wednesday, 29 October 2025, 11:37 AM**Time taken** 16 mins 12 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int max(int a, int b) {
4      return (a > b) ? a : b;
5  }
6
7  int main() {
8      int n;
9      scanf("%d", &n);
10
11     int seq[n], dp[n];
12
13     for (int i = 0; i < n; i++) {
14         scanf("%d", &seq[i]);
15         dp[i] = 1; // every element is a subsequence of length 1
16     }
17
18     for (int i = 1; i < n; i++) {
19         for (int j = 0; j < i; j++) {
20             if (seq[j] <= seq[i]) {
21                 dp[i] = max(dp[i], dp[j] + 1);
22             }
23         }
24     }
25
26     int result = 0;
27     for (int i = 0; i < n; i++) {
28         result = max(result, dp[i]);
29     }
30
31     printf("%d\n", result);
32     return 0;
33 }
```

	Input	Expected	Got	
✓	9 -1 3 4 5 2 2 2 2 3	6	6	✓
✓	7 1 2 2 4 5 7 6	6	6	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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